From Sensor Streams to Bounded Machine Memory: Architecture, Negative Transfer Results, and Domain Specialists

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ABSTRACT

Industrial and edge equipment produce continuous telemetry that cannot be kept in full on flash storage, while first-in-first-out buffers overwrite rare fault episodes with routine data. We study µEngram, a bounded stateful memory that compresses short sensor windows into noise-whitened low-dimensional vectors, tracks operating regimes with a fixed budget of semantic, candidate and episodic slots, quarantines unverified shifts, and pins raw waveform slices for rare episodes. We evaluate it and several encoder variants on a synthetic generator and ten public benchmarks: CMAPSS, MetroPT-2 and MetroPT-3, SKAB, SCANIA Component X, OS-Metrics, EngineAD, MFPT, CWRU and TSB-AD-U. On the synthetic stream the memory cuts write operations by 86.2% over 10 seeds while holding every injected event, but this is a modeled count, not measured flash wear. On public data, ranking quality and operational early warning diverge. Frozen detectors give no pre-report signal inside MetroPT-3's uncommissioned load state, catch 5 of 34 SKAB faults at the highest false-alarm rate of three compared streaming methods, and score near chance on held-out EngineAD trucks. Distilling foundation models into a 46 KB student improves window ranking on some tasks, yet every model flags the held-out healthy recording in both bearing tests, and the students raise CMAPSS alarms earlier than the maintenance horizon. A role-separated, per-subsystem architecture with pre-fit numerical diagnostics detects 29 of 30 test-split events on one OS-Metrics workload, against 27 of 30 for a single-encoder baseline, with fewer false alarms. All results come from single-workload studies without a field validation cohort.

PRINCIPAL FINDINGS

1. **Bounded memory works where its assumptions hold:** 86.2% fewer write operations with every injected event retained over 10 synthetic seeds. Bytes written rise 37% because raw windows are pinned.
2. **Ranking is not warning:** every CWRU model reaches 1.0000 AUROC, yet every model flags the held-out healthy recording (normal-file false-alarm rate 1.0).
3. **Proxy screens mislead:** the label-free screen's winner (middle layer, score 0.610) has a lower event median than the bottleneck in the full alarm pipeline on both SKAB and MetroPT-3.
4. **Distillation shifts alarm timing:** on CMAPSS all four models alarm before failure on 10 of 10 engines, but the students' alarms start earlier and more often than the ordinary CNN's.
5. **Partitioning beats pooling on one workload:** domain specialists detect 29 of 30 OS-Metrics test events at 19.6 false alarms per week, against 27 of 30 at 26.6 for the DSP baseline.
6. **Detector state must survive restarts:** after a simulated restart with no detector checkpoint, the engine re-emitted lifecycle records and lost alarm-open records in 120 of 120 Lite-only and sidecar runs, and in 406 of 448 machine-and-restart-position cells across 28 server machines where the uninterrupted run alarms. A SQLite input log plus engine checkpoint matched the uninterrupted run in all 60 runs, all 448 cells and all 780 SIGKILL trials, including against the real EdgeMemory Lite library. Durability is not free: full synchronization costs 8 to 490 ms per window on real disks, against 1.5 to 2.9 ms with relaxed synchronization whose power-loss durability is untested (Section 5.14).
7. **Modeled crash images expose a durability trade-off:** across 6,599 whole-syscall crash images, `FULL` and `NORMAL` recovered a consistent capsule and final vendor state, but `NORMAL` lost acknowledged input windows in 1,938 images; a no-journal control corrupted 166 of 483. This is not hardware power-loss evidence.

1. INTRODUCTION

Embedded telemetry systems face a structural retention dilemma. At one 16-float vector per second (86,400 vectors per day), raw vector payloads alone consume about 5.3 MiB daily, before timestamps, metadata, index structures and high-rate source waveforms. Updating approximate nearest-neighbor graph indexes such as HNSW in place is expected to introduce random-write amplification on raw NAND flash and eMMC storage (not measured here). Ring buffers that evict strictly by age discard historical fault signatures whenever a machine runs normally for days after an incident.

This work began as a proposal to port cloud memory frameworks for language-model agents, specifically Mem0 [1], to edge devices using vector search. Testing that idea against physical telemetry exposed three mismatches:

1. **Physical state is not conversational memory.** Mining trucks, wind turbines and compressors do not need entity extraction to remember user preferences. They need bounded tracking of wear baselines, operating regimes and acute fault excursions.
2. **The ingestion path must be deterministic.** Cloud Mem0 invokes a language model on every write. On a gateway ingesting 100 Hz to 1 kHz sensor streams, a language model in the write loop violates real-time latency and memory bounds. Large models also strain constrained devices [2]. Ingestion and consolidation must remain deterministic and mathematical.
3. **Memory and flash wear need explicit bounds.** Continuous mutation of in-memory graph indexes is expected to starve constrained RAM and accelerate flash wear (not measured here; see the hardware gate in Section 6). Edge state memory needs tiered, density-aware compaction and strict admission gates.

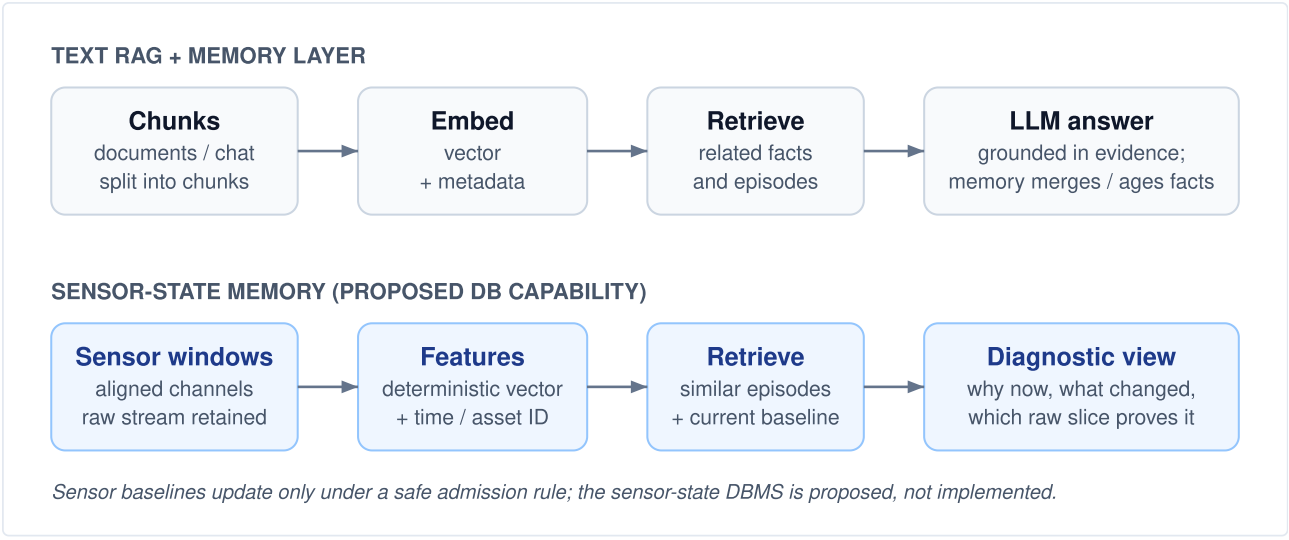


Figure 1. Text retrieval-augmented generation versus the proposed sensor-state memory pipeline: two kinds of memory, one lifecycle idea.

Figure 1 contrasts the two designs. We analyze a Python reference implementation of the sensor-state design, µEngram, together with an embedded event-store adapter, across synthetic and public datasets.

Contributions.

- A bounded memory architecture for sensor windows: noise-whitened projection, a fixed budget of semantic, candidate and episodic slots with competitive admission, a pin-before-write persistence layer, and a label-free alarm policy (Section 3).
- A cross-benchmark evaluation that separates mechanism results from operational early-warning results, with explicit definitions of every metric and split (Sections 4 and 5).
- Negative and diagnostic findings that are easy to miss: wide embeddings that regress, proxy screens that mislead, foundation-model students that flag healthy data, and alarm-tuning sweeps that cannot meet all operational budgets (Section 5).
- A role-separated, per-subsystem architecture with pre-fit numerical diagnostics, evaluated on one workload (Sections 3.6, 3.7 and 5.9).
- A host-owned durable capsule (an input write-ahead log and a versioned engine checkpoint in one SQLite store, with a configurable checkpoint interval and synchronization mode) and a restart study that measures what losing detector state costs, whether recovery survives process death, and what durability costs on real storage (Sections 3.5 and 5.14).

The remainder of the paper reviews related work (Section 2), describes the method (Section 3), the experimental setup (Section 4) and the results (Section 5), and closes with limitations (Section 6) and conclusions (Section 7).

2. RELATED WORK

Fault detection and noise whitening. Model-based failure detection relies on analytical redundancy, comparing a system's behavior with a model of normal operation [3], [4]. For multichannel data, the maximum noise fraction transform orders components by signal-to-noise ratio, which motivates estimating a noise covariance and whitening with it [5]. We estimate noise from lagged differences inside a single operating state and use it to whiten a PCA projection.

Bounded streaming memory. MemStream uses a thresholded memory of recent normal items for streaming anomaly detection [6]. Continual-learning work addresses similar admission and eviction problems with diversity-aware replay [7], constrained buffer allocation [8] and prototype replay [9]. MIXAD relates memory to

explainable time-series anomaly detection [10]. Our slot memory differs by separating semantic, candidate and episodic roles and by admitting rare episodes only when their importance exceeds that of the weakest stored episode.

Time-series foundation models and representation depth. MOMENT [11] and Chronos-2 [12] provide pretrained time-series representations. Intermediate layers of such models can carry localized anomaly signals, as studied in TimeRep [13], LayerIM [14], PaAno [15] and MOMEMTO [16]. Deploying large models on edge hardware remains a constraint [2]. We distill two foundation models into a small student and test whether intermediate readouts help, using alarm-level rather than ranking-only criteria.

Benchmarks and evaluation. Recent benchmark suites stress careful protocol design for time-series anomaly detection [17], [18], and foundation-model detectors have been proposed [19]. We follow that spirit by reporting ranking metrics (AUROC, average precision) next to event-level and false-alarm metrics, and by recording where labels enter fitting.

3. METHOD

3.1 Problem Setting

A sensor stream is segmented into causal windows $X \in \mathbb{R}^{C \times K}$ with C channels and K samples. A window at step t uses only samples up to t. A memory with a fixed budget of B slots must (i) decide per window whether it is routine, a new state or a rare excursion, (ii) decide what to keep and what to write to flash, and (iii) raise an alarm when a window is anomalous in a way that persists. Windows are processed one at a time, deterministically, with no language model in the loop.

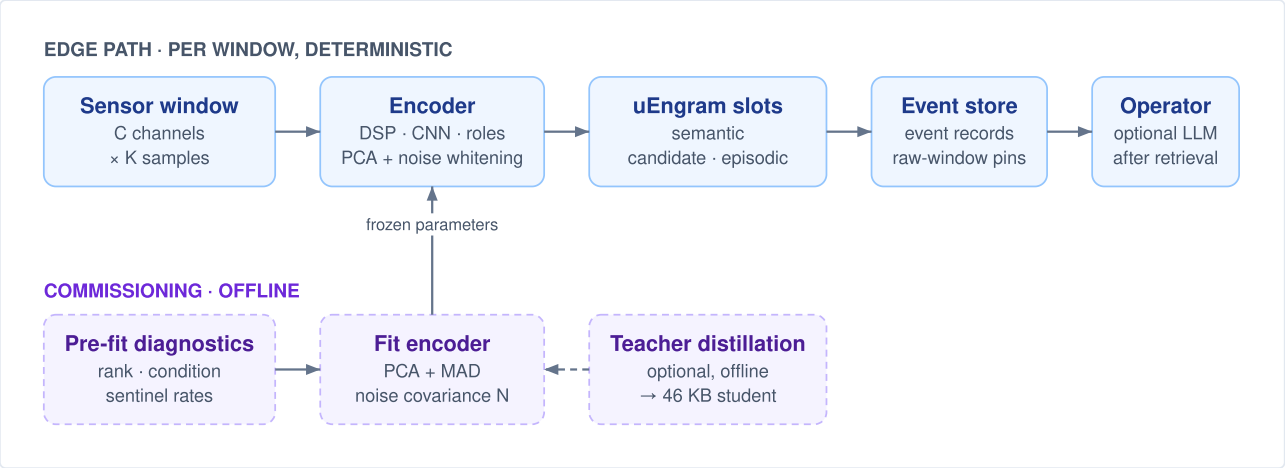


Figure 2. µEngram edge path (per window, deterministic) and offline commissioning path.

Figure 2 shows the edge path and the offline commissioning path that fits the encoder and noise covariance on healthy data.

3.2 Encoders and Noise-Whitened Projection

DSP encoder. For each channel the baseline encoder computes six features: the mean, the log standard deviation, and the log energy in each of four equal-width bands of the positive-frequency power spectrum (the DC bin is excluded). This yields $6C$ features, which are median-and-interquartile-range scaled and projected by principal component analysis to dimension d (default $d = 16$).

Noise whitening. Because raw sensor features covary under load changes, Euclidean distance overweights correlated channels. We estimate within-state noise from differences of windows L steps apart inside the same operating group [5]. Algorithm 1 gives the estimate: a trimmed, reweighted covariance of the differences ignores

jumps caused by regime switches and faults. For a representation $r(x)$ with mean $\bar{r}$, PCA eigenvectors V, eigenvalues Λ, and noise covariance N, the whitened state vector is

$$z(x) = N^{-1/2}\Lambda^{-1/2}V^{\top}(r(x) - \bar{r}).$$

Eigenvalues of N below a small multiple of the largest are floored before inversion. Setting d too wide introduces collinear frequency bands that make N ill-conditioned or singular (Section 3.7).

Algorithm 1. Robust within-state noise covariance.

```
Input: projected training vectors z₁ … zₙ, lag L, trim quantile q_t = 0.975, iterations I = 5
Output: noise covariance N
D ← { (z_{t+L} − z_t) / √2 : t and t+L in the same group }
q ← x²_d quantile at q_t ;  c ← q_t / P(x²_{d+2} ≤ q)          ▷ consistency factor for trimming
N ← diag(σ²),  σ̂_j ← 1.4826 · MAD(D_j)                        ▷ robust per-dimension start
repeat I times
    keep ← { Δ ∈ D : Δᵀ N⁻¹ Δ ≤ q }                            ▷ drops regime jumps and steps
    if |keep| < max(10, d + 1) then raise an error
    N ← c · Cov(keep)
return N
```

CNN encoder. The alternative encoder is a two-stage one-dimensional convolutional autoencoder. Two convolutions (32 and 64 filters, kernel 5, stride 2, GELU) feed a temporal average pool and a linear projection to the latent dimension d. A decoder reconstructs the window. Training minimizes reconstruction error, optionally with an orthogonality penalty on the projection, using AdamW (learning rate 10^{-3}, weight decay 10^{-4}). The latent vector is then whitened as above. For one input channel and $d = 16$ the encoder holds 11,536 parameters, 46,144 bytes in float32. Three readouts are used in Section 5.7: the projected bottleneck, a pooled first-convolution map ("middle layer"), and the concatenation of pooled first- and second-convolution maps ("multi-level").

3.3 Bounded State Slots and Competitive Admission

The memory budget is divided into **semantic slots** (compact summaries of confirmed routine operating modes), **candidate slots** (unfamiliar patterns on persistence probation) and **episodic slots** (rare excursions paired with a pointer to a raw time-series slice). Each slot keeps a running mean, a predictive variance and a hit count. A window lies inside a slot's gate when its diagonal Mahalanobis distance is below a χ^2 threshold at level 10^{-4}.

Early designs evicted the weakest mature episodic slot whenever any out-of-gate sample arrived, which let mild outliers overwrite severe historical faults. We therefore use **competitive rarity admission** (Figure 3). An out-of-gate window enters a full episodic bank only if its surprisal exceeds the importance of the weakest stored episode,

$$\text{importance} = \text{surprisal} \times (1 + \ln n) \times 2^{-\text{age}/h},$$

where n is the episode's hit count and h the episodic half-life; otherwise it is discarded, while probation candidates remain replaceable. The design draws on thresholded streaming memory [6], diversity-aware replay [7], constrained buffer allocation [8] and prototype replay [9].

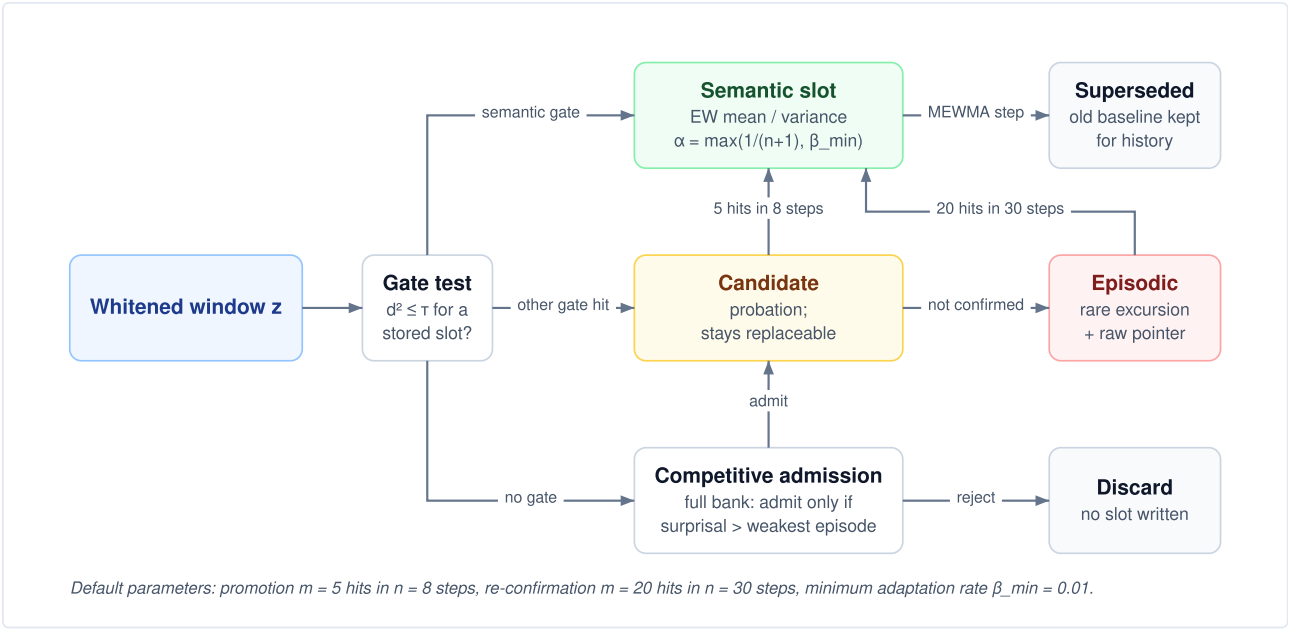


Figure 3. Slot state machine: gate routing, candidate probation, promotion, step supersession, and competitive episodic admission. A window inside an episodic slot's gate is absorbed by that slot; only the candidate path is drawn.

Algorithm 2 is the per-window update. Novelty is scored against semantic (normal) slots only, so a recurring fault held in an episodic slot cannot mask itself. Rarity, the distance to any stored slot, ranks what is worth keeping. A multivariate exponentially weighted moving average of standardized residuals detects step changes in a semantic slot's level, supersedes the old slot and opens a new one. We ablate competitive admission in Section 5.1.

Algorithm 2. Per-window slot lifecycle.

```
   Input: whitened vector z, step t, raw-window index r
G ← active candidate, episodic and semantic slots
for each k in G: d²_k ← Σ_j (z_j − μ_kj)² / v_kj                ▷ v: predictive variance
score ← min over semantic k of d²_k ;  rarity ← min over G of d²_k
alarm ← score > τ_alarm
if some d²_k ≤ τ then                                          ▷ window is inside a gate
    P ← semantic slots inside τ if any, else all slots inside τ
    k* ← argmin over P of d²_k + Σ_j log v_kj
    ABSORB(k*, z, t)
else
    if not MAKEEPISODICROOM(t, surprisal(rarity)) then return discard
    open a candidate slot at z with raw pointer r               ▷ action: new candidate
candidates older than n_promote steps become episodic
every consolidate_every steps: merge duplicate semantic slots, checkpoint drift, supersede dormant ones
   Procedure ABSORB(k, z, t)
α ← max(1/(n_k+1), β_min) if k is semantic, else 1/(n_k+1)
e ← (z − μ_k) / √v_k ;  update (μ_k, S_k) with rate α ;  n_k ← n_k + 1
if k is semantic then
    E_k ← λ·e / s_step + (1 − λ)·E_k                            ▷ MEWMA, λ = 0.1
    if n_k ≥ n_min and c·‖E_k‖² > h then split k; old slot superseded   ▷ step change
    return
record hit t ;  (m, N) ← (5, 8) for a candidate, (20, 30) for an episodic slot
if the m-th latest hit is within N steps and semantic room exists then promote k   ▷ promotion
```

3.4 Alarm Policy

Each window yields two scores. **Novelty** is the minimum diagonal Mahalanobis distance to semantic slots. **Health** is the minimum distance to a frozen healthy reference set captured at commissioning. Thresholds are calibrated without labels on a healthy calibration split by the conformal upper quantile: the threshold is the

$\lceil (1-\alpha)(m+1) \rceil$-th order statistic of m calibration scores. Thresholds are set per operating state when that state has at least 100 calibration windows, and globally otherwise.

A persistence counter rises by one on each threshold exceedance, falls by one otherwise, saturates at k, and opens an alarm when it reaches k. The alarm closes when the counter clears. Windows that cannot be monitored (the first window of a segment, or the first after a data gap) leave the counter unchanged. A window whose nearest slot is a fresh candidate and whose health is normal raises a separate *new-state* alert instead of a novelty alarm. An optional external-evidence channel can corroborate alarms with a three-valued AND; it is not used in any reported experiment.

The configuration used throughout was selected on MetroPT-3 development data and then frozen: $\alpha = 0.003$, novelty persistence $k = 3$, the health channel disabled, per-state thresholds, and the new-state alert enabled. It is applied unchanged to the other datasets, so MetroPT-3 results under it are development results, not confirmation.

3.5 Layered Persistence

The reference adapter (Figure 4, Table 1) keeps live states in memory and writes versioned event records to an embedded event store: new candidates, promotions, step changes, alarm open and clear transitions, and any window that produced a lifecycle event (merge, eviction or supersession). Routine windows are not written. A raw window is pinned before its event record is written, and the pin is released if the write fails, so a retry never re-feeds the window into the engine.

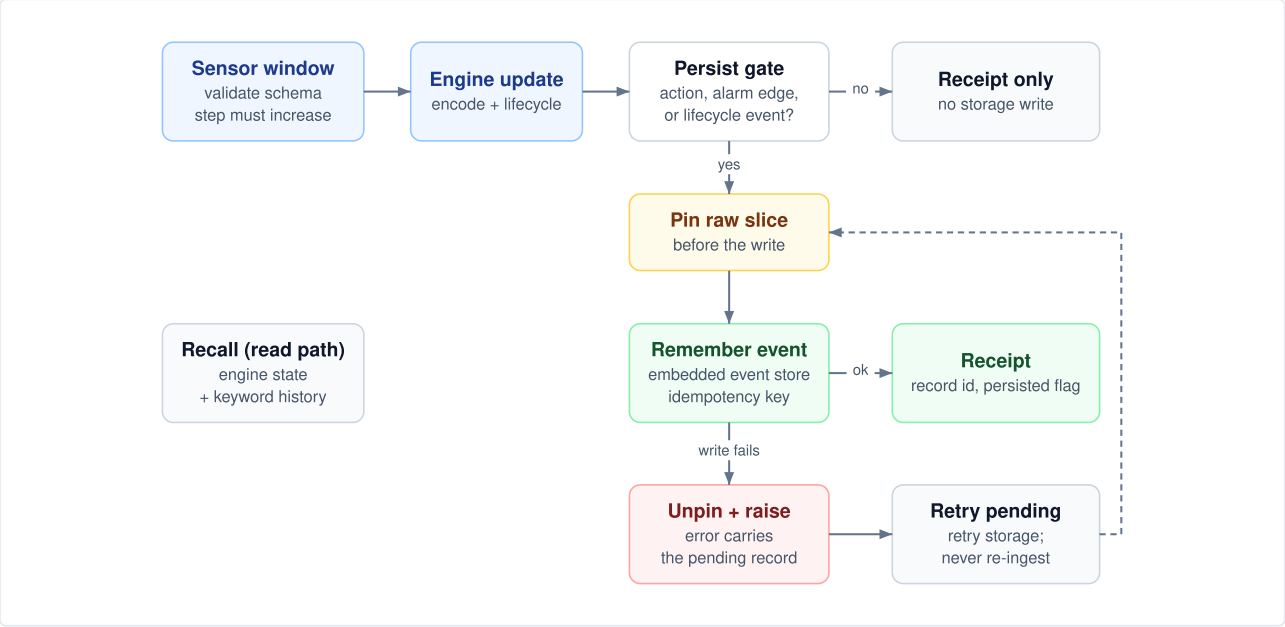


Figure 4. Sensor-state persistence pipeline in the implemented adapter. The raw window is pinned before the event write; on failure the pin is released and the error carries the pending record for retry.

Table 1. Adapter stages: interface and behavior when the stage fails.

Stage	Interface	Failure behavior
Window validation	Validated window record	Rejected with an error before the engine consumes it
Encoding and whitening	Host-selected encoder	Outside the adapter's error handling
Live bounded memory	Ingest, recall, drain lifecycle events	Window already consumed; recovery retries storage and never re-ingests
Persist gate	Internal decision	Returns a receipt with no storage write

Stage	Interface	Failure behavior
Raw-window pin	Pin and unpin by reference and owner	Pin released if the event write fails, then the error is raised
Event storage	External embedded store: remember, recall, forget	A persistence error carries the pending records for retry
Retrieval bundle	Exact state recall plus keyword history	Invalid mode or non-finite query rejected
Alarm combination	Combination rule over channels	Not on the write path; only flagged transitions are persisted

Detector checkpoint and input log. The adapter's outbox makes event *delivery* restart-safe but does not preserve detector state. A single-host SQLite store, the *capsule*, adds three things in write-ahead-log mode, with full synchronous commits by default: an input log written before the engine consumes a window, a versioned checkpoint of the engine arrays and alarm state, and the outbox with an exact vector index. Recovery restores the latest checkpoint and replays only logged windows newer than it; stable idempotency keys make the vendor-side replay safe. The checkpoint interval and the SQLite synchronization mode are configurable, and Section 5.14 measures their cost and tests process-death safety. Checkpoints are NumPy archives with no executable deserialization and are refused when the engine configuration, dimension, array shapes or dtypes differ. The capsule is host-local: it does not turn the embedded event store and the capsule into one transaction (Section 6).

3.6 Role-Separated Encoders and Domain Specialists

Monolithic encoders mix incompatible physical variations into one distance metric, which plausibly explains their failure on multichannel telemetry (untested hypothesis). We therefore decompose telemetry into four branches:

- **Mode:** distance to known operating-context centers, kept as diagnostic context and excluded from alarm fusion.
- **Level:** per-channel window means.
- **Dynamic:** mean and root-mean-square of lagged sample differences at lags 1, 2, 4 and 8, which capture multiscale frequency and transient shifts.
- **Health:** distance from a frozen commissioning reference.

Level and dynamic features are scaled robustly (median and interquartile range), then projected and noise-whitened as independent blocks. Each alarm branch score s becomes an upper-tail empirical p-value against n scores $c_1 \dots c_n$ from a separate healthy calibration stream,

$$p(s) = \frac{1 + \#\{\, i : c_i \geq s \,\}}{n + 1},$$

and the branch fires when $p \leq \alpha$ (here $\alpha = 0.01$). Fusion policies combine the level, dynamic and health branches: any branch, at least two, dynamic and health together, or all. An alarm opens after three consecutive fired windows.

A **domain-specialist architecture** assigns each subsystem its own role-separated encoder over a named channel group, so that localized spikes are not diluted by quiet channels (Algorithm 3). A specialist fires under the dynamic-and-health policy, the fused alarm is the OR over specialists, and the set of firing specialists gives a root-cause attribution.

Algorithm 3. Domain-specialist scoring, fusion and routing.

```
Input: window x; specialists s with channel sets C_s (for example CPU, memory, network); level α = 0.01
Output: alarm, triggering specialists A, memory route
for each specialist s do
    p_s,b ← calibrated p-value of branch b ∈ {level, dynamic, health} on x[C_s]
    flag_s ← (p_s,dynamic ≤ α) and (p_s,health ≤ α)              ▷ dynamic-and-health branch policy
alarm ← flag_CPU or flag_Memory or flag_Network                  ▷ OR over specialists
A ← { s : flag_s }                                               ▷ root-cause attribution
score ← Σ_s -log min_b p_s,b                                     ▷ continuous fused score
if data quality fails then route ← QUARANTINE
else if alarm is open or A ≠ ∅ then route ← EPISODIC (with A)
else if operator confirmed normal then route ← SEMANTIC
else route ← QUARANTINE
```

Only operator-confirmed normal windows enter semantic memory. Unconfirmed normal windows route to quarantine, and alarmed windows route to episodic memory.

3.7 Pre-Fit Numerical Diagnostics

Before fitting, a diagnostic inspects the channel covariance for effective rank, condition number, sentinel rates and null-space collinearity. The effective rank counts eigenvalues above $\epsilon\,\lambda_{\max}\,C$, where ϵ is machine epsilon. The status is *singular* when the rank falls below the channel count, *ill-conditioned* when the condition number exceeds 10^{12}, *moderately conditioned* above 10^{6}, and *well-conditioned* otherwise. For a singular matrix, null-space vectors identify the redundant channel groups. The diagnostic exists to turn a silent numerical failure into an explicit, explainable error before fitting.

Fail-soft estimation. The strict noise and step-inflation estimators still raise when the data cannot support them, but the encoders no longer stop there. A labeled ladder steps down (noise: robust, ungrouped, shrunk; step inflation: exact, group-segmented, assumed white), emits a warning and records the rung in the result row; turning warnings into errors restores the strict behavior. The step-inflation estimate must also rest on at least 100 warm lag pairs (relative standard error about $\sqrt{2/n}$, 14%). Below that, it uses the unit groups when they give more pairs. Steady degradation trips the jump detector and fragments each CMAPSS unit, so the strict estimate had 40, 17 and 14 pairs on FD001 $d = 24$, FD002 (normalized) and FD003, came out below 1 and was clamped to white noise (Section 5.3 reports the effect).

3.8 Offline Teacher Distillation

Deploying large time-series foundation models on gateways violates memory and latency constraints [2] (the two teachers used here have 35.3 M and 119.5 M parameters), so we distill them offline. During commissioning, causal windows X_t are passed to one or M frozen teachers (MOMENT-1-small [11] with 512-dimensional embeddings, and Chronos-2 [12] with a 32-dimensional quantile feature vector for one channel) while a small student CNN S_θ minimizes reconstruction error alongside dimension-normalized feature matching:

$$\mathcal{L}(\theta) = \mathrm{MSE}\left(D(S_\theta(X_t)), X_t\right) + \lambda \frac{1}{B}\sum_{b=1}^{B} \mathrm{MSE}_b\left(H(S_\theta(X_t)), T(X_t)\right) + \lambda_{\mathrm{orth}}\,\mathrm{mean}\left((WW^\top - I)^2\right).$$

Here D is the decoder, H a linear teacher head, T the standardized teacher features, B the number of teacher feature blocks, MSE_b the error on the b-th block, and $\lambda = 0.25$. Averaging per block stops a 514-dimensional block from outweighing a 32-dimensional one. Teachers are used only in training; the deployed student is the 46,144-byte encoder of Section 3.2.

Because foundation models are often applied after per-window normalization, which erases absolute pressure or vibration levels, we append raw per-channel statistics to the teacher embedding, $T^{+}(X) = [T(X), \mu_1, \ldots, \mu_C, \sigma_1, \ldots, \sigma_C]$ (per-window mean and standard deviation). MOMENT's embedding path

normalizes each window internally, so the appended statistics restore the level information. When a window is shorter than the teacher's context, we compare left-edge padding with linear resampling.

4. EXPERIMENTAL SETUP

4.1 Datasets

Table 2 summarizes the datasets. All public datasets are used as released. Dataset sources are cited in the table and listed under Data and Code Availability.

Table 2. Datasets, windowing and splits.

Dataset	Channels	Window	Fit / calibration / test	Events and labels
Synthetic engine stream	6	$K = 64$	8,000-window fit stream; 4,000 fault-free calibration windows; 20,000 evaluation windows	Three anomaly families (knock, cavitation, thermal) plus three one-window seizure events at 25%, 50% and 85% of the stream; three operating regimes
CMAPSS FD001 [20]	14 sensors	$K = 16$ cycles, stride 1	Encoder fit on engines 1–50; commissioning on the first 30 windows of each; scored stream is engines 51–100	Late life: remaining useful life (RUL) ≤ 20; healthy: RUL > 75; windows between are excluded from ranking metrics
MetroPT-3 [21]	7 analog	$K = 60$ (about 10 min)	Fit before 22 Feb 2020; calibration to 1 Mar 2020; test after	4 operator-annotated air-leak intervals
MetroPT-2 [22], [23]	7 analog plus context	$K = 60$ after 10 s decimation	January: fit to 22 Jan, calibration to 29 Jan 2022. April: fit to 19 May, calibration to 26 May 2022	3 January and 2 April events
SKAB [24]	8	$K = 10$ (10 s), non-overlapping	First 30% of one anomaly-free file (282 windows) fits; the remaining 658 windows calibrate; 34 fault files test	One contiguous fault interval per file (34 events)
SCANIA Component X [25]	105 features	One row per window	Healthy training vehicles split by identifier modulo 5; all validation rows test	Five repair-proximity classes over 5,046 validation vehicles
OS-Metrics [26]	5	$K = 16$ (4 min)	Day 1 fits; day 2 calibrates; days 3–7 test; windows overlapping events are removed from fit and calibration	42 annotated disturbances (13 CPU, 12 memory, 17 network)
MFPT [27]	1	$K = 512$ at 2,048 Hz, stride 128	Two normal recordings fit; one held-out normal recording and 20 fault recordings test	Per-recording fault or normal
CWRU [28]	1 (drive end)	$K = 512$ at 12 kHz, stride 256	Normal recordings at 0–2 HP fit; one held-out normal recording at 3 HP and 12 fault recordings test	Inner race, rolling element, outer race
TSB-AD-U [17]	1 (six series)	$K = 64$, stride 32	First half of each series' pre-anomaly prefix fits; later windows test	Window positive if any point is anomalous
EngineAD [29]	8 (released principal components)	300-point segments, as released	15 trucks fit (normal-only segments, 150,442); 5 trucks calibrate (41,671 normal segments); 5 held-out trucks test (55,890 segments)	Segment label = first point label (9,810 anomalous test segments, 17.6%); 135 of 269,247 segments have mixed point labels

Dataset	Channels	Window	Fit / calibration / test	Events and labels
SMD [30]	38	$K = 10$ (10 min at 1-minute sampling), non-overlapping	Per machine: the first 70% of the normal-only training file fits and the last 30% calibrates; the labelled test file is scored; 28 machines	Maximal runs of anomalous test rows (327 events over 28 machines)

Windows are causal and, except where a stride is stated, non-overlapping and gap-aware: no window spans a data gap or a split boundary, and the first window after a gap is not monitorable. MetroPT-3 and MetroPT-2 clean negatives exclude the 24 hours before and 48 hours after each event. SKAB uses no exclusion buffer. OS-Metrics removes fit and calibration windows that overlap an annotated event, so event labels enter fitting there.

4.2 Metrics

- **Event retention.** On the synthetic stream, the fraction of distinct labeled events represented by at least one live memory item at the end of the run. It measures what the memory keeps, not detection.
- **Write reduction.** $100\,(1 - W/T)$, where W is the number of physical write operations and T the number of ingested windows. It can be negative. **Bytes written** counts payload bytes, including slot flushes and 16-byte deletion tombstones.
- **AUROC and average precision (AP).** Window-level, over windows labeled event or clean negative.
- **Event detection.** Two criteria are used and each result states which. *Strict* detection requires a persistent alarm to start inside the event interval; two-hour recall also accepts a start up to two hours before onset. *Active-in-interval* detection, used by the calibrated alarm pipeline, counts an event whenever an alarm is active at any time inside its interval, including one that began earlier. Active-in-interval is the more lenient criterion, so counts under the two are not comparable.
- **False alarms per week.** Alarm-opening episodes on clean negatives, with openings less than one hour apart merged, divided by the observed clean exposure in weeks. Where the clean exposure is a few hours, weekly rates are extrapolations. **Alarm-time share** is the fraction of observed clean time spent in alarm.
- **Lead time.** First alarm time minus event start, so a negative value means the alarm came before the event.
- **CMAPSS horizon metrics.** An alarm episode is a run of three consecutive threshold exceedances. *Recall before failure* counts engines with at least one episode. *Recall at the horizon* counts engines with an episode starting at RUL $\leq$ 20. *Pre-horizon episodes* count episode starts at RUL > 20.
- **RUL RMSE.** Root mean squared error over test engines. We report a memory-lookup predictor and a health-score regressor.
- **File-level metrics (bearings).** *File AUROC* is the AUROC of each recording's maximum window score against the file's fault label. The *normal-file false-alarm rate* is the fraction of held-out normal recordings whose maximum score exceeds the 99th percentile of training-normal scores. *Pooled AUROC* pools all windows of all series; *equal-series AUROC* is the mean of the per-series AUROCs.

4.3 Baselines, Seeds and Reporting

On the synthetic stream we compare: **naive** (append every vector without limit), **FIFO** (circular buffer of B vectors), **reservoir** sampling, a **centroid-clustering store** in a label-free variant and an oracle variant that receives ground-truth severity, µEngram, and three µEngram ablations (fixed slot variance, no candidate promotion, and no step detection). The budget is $B = 256$ slots in every experiment. The synthetic and CMAPSS lifecycle experiments assign 81.25% of them to episodic slots (208 episodic, 48 semantic); the field alarm experiments (MetroPT, SKAB, SCANIA) assign a quarter (64 episodic, 192 semantic). Other defaults are in Table 3. The synthetic experiments use seeds 0–9. The CMAPSS lifecycle experiment uses seeds 0–4, MetroPT-3 uses five seeds, EngineAD uses ten seeds at each of three encoder widths, and SKAB, SCANIA,

MetroPT-2 and the foundation-model gates use a single seed. We report means with ranges across seeds; where a result is deterministic across seeds, we say so. Single-seed results are single-run estimates with no variance estimate.

Table 3. Default slot-engine parameters.

Parameter	Value	Parameter	Value
Semantic / episodic slots ($B = 256$)	48 / 208 (synthetic, CMAPSS); 192 / 64 (field alarm runs)	Gate level	10^{-4}
Minimum adaptation rate $\beta_{\min}$	0.01	Candidate promotion	5 hits in 8 steps
Re-confirmation of episodic slot	20 hits in 30 steps	Consolidation period	200 steps
Duplicate-merge Bhattacharyya threshold	0.5	Dormant age	2,000 steps
Semantic half-life	2,000 steps	Episodic half-life	20,000 steps
Step detector λ / level	0.1 / 10^{-6}	Ledger flush period	50 writes

5. RESULTS

5.1 Synthetic Stream: Lifecycle Against Controls

On the synthetic stream with the DSP encoder at $d = 16$, µEngram retains every injected event in every seed (retention 1.00) and cuts write operations by 86.2% on average (95% bootstrap interval over seeds 85.9–86.5%; range 85.3–87.2%; about 2,762 of 20,000 windows written). Average precision is 0.9955 (interval 0.991–0.999; range 0.9795–1.0000). Table 4 compares the controls. Intervals in this section are percentile bootstrap intervals of the mean over the 10 seeds (10,000 resamples); they quantify seed-to-seed variation of one generator, not uncertainty about other streams.

Table 4. Synthetic stream, 10 seeds, means. Event retention (Ret.), write-operation reduction (W.red., %) and AP at $d = 16$ and $d = 36$.

Method	Ret. (d=16)	W.red.	AP	Ret. (d=36)	W.red.	AP
Naive	1.000	0.0	0.465	1.000	0.0	0.446
FIFO	0.011	0.0	0.531	0.011	0.0	0.304
Reservoir	0.022	93.2	0.857	0.022	93.2	0.581
Centroid store, label-free	0.542	14.2	0.023	1.000	93.1	0.137
Centroid store, oracle severity	0.294	10.3	0.018	0.787	90.3	0.042
µEngram	**1.000**	86.2	**0.996**	0.801	71.6	0.534
µEngram, fixed variance	1.000	84.7	0.999	0.651	74.1	0.493
µEngram, no promotion	1.000	–9.0	0.114	0.745	–7.8	0.298

Three observations follow. First, only µEngram and the naive store retain every event at $d = 16$; FIFO and reservoir sampling discard nearly all of them while saving writes (reservoir) or not (FIFO). Second, candidate promotion is essential to the write saving: without it, writes exceed ingested windows (–9.0%) and bytes written rise to 31.0 MB. Third, the write count is a benchmark model. Because µEngram pins raw windows, it

writes 2.19 MB against 1.60 MB for the naive store, 37% more bytes, even though it performs 86.2% fewer write operations. The CNN-encoder variant retained only 0.262 of events despite high AUROC. This stream is generated under assumptions we control, so the result validates mechanism, not field behavior.

Competitive admission ablation. We compared competitive admission (Section 3.3) with accepting every arrival into a full episodic bank, at budgets of 2, 4, 8, 16 and 256 slots (DSP encoder, 5 seeds). At the budgets where the bank saturates, competition retains far more events: retention is 0.043, 0.190, 0.535 and 0.901 at budgets 2, 4, 8 and 16, against 0.001, 0.010, 0.274 and 0.778 for accept-all. Write reduction is also higher at budgets 2 to 8 (27.1, 42.8 and 90.1% against −7.1, 22.1 and 81.5%) and similar at 16 (86.7 against 85.9%). The cost is ranking at one budget: AP at budget 8 is 0.428 with competition against 0.614 for accept-all, while at budget 16 the two are close (0.982 and 0.985). AP is undefined at budget 2 for both. At the synthetic experiments' budget of 256 the episodic bank never saturates within 20,000 windows, so the two rules are identical there. We did not test whether the field experiments, which use a smaller episodic bank, saturate it.

5.2 Encoder Width Regressions

Widening the DSP projection from $d = 16$ to $d = 36$ lowers µEngram's event retention from 1.00 to 0.801 (interval 0.784–0.819; range 0.766–0.845 over seeds) and its AP from 0.996 to 0.534 (interval 0.483–0.586; median 0.541). Paired by seed, the change is −0.199 in retention (interval −0.216 to −0.181), −0.462 in AP (−0.510 to −0.411) and −14.6 points of write reduction (−15.0 to −14.2), so the regression is not a seed artifact. Bytes written rise to 4.79 MB, 50% above the naive store. At $d = 36$ the label-free centroid store retains every event with a 93.1% write reduction but an AP of 0.137 (interval 0.118–0.157), so no method dominates on all three measures. On the 8-channel SKAB benchmark the effect is mixed (Table 6, Section 5.4): $d = 48$ detects more events but spends 18.6% of clean time in alarm. The sweeps do not identify an embedding dimension that is best across datasets.

Whitening by lag-1 noise treats slow within-state variation as novelty. A diagnostic on one seed's training latents (not committed) suggests that at $d = 36$ most dimensions carry it: the lag-100 difference noise exceeds the lag-1 noise by more than a factor of 2 in 26 of 36 dimensions, against 4 of 16 at $d = 16$. Setting each dimension's gate scale σ_0 to its lag-20 difference noise when that is larger raises µEngram's event retention at $d = 36$ from 0.801 to 0.999 (range 0.989–1.000 over seeds) and its AP from 0.534 to 0.591 (median 0.605), with 74.7% write reduction (default 71.6%) and 194 live items (default 256). AP stays far below its $d = 16$ value of 0.996, so the option restores retention but not precision. It is off by default and the published tables use the default; its effect at other widths and datasets was checked only exploratorily, with mixed results (for example SKAB $d = 48$ detected 7 of 34 events against 9), and that check is not committed.

5.3 Core Benchmark Evidence

Table 5 summarizes the experiments on field-like data, and Figure 5 shows the per-seed spread of the EngineAD result. They expose a sharp divide between point-wise ranking and operational early warning.

Table 5. Core benchmark evidence.

Dataset	Configuration	Primary result	Interpretation
CMAPSS FD001 (5 seeds)	DSP, $d = 16$	Health AUROC 0.999, AP 0.995; RUL RMSE 42.1 (memory lookup), 25.5 (health score) and 23.5 (health score and unit age) against 25.8 for naive storage (range 23.2–25.8)	Separating early from late life does not yield remaining-life regression better than the naive baseline. $d = 32$ earlier failed with "fewer than 10 difference pairs for noise estimate"; it now runs through the fail-soft estimators (unit-grouped step inflation) with health AUROC 0.998, end-of-life retention 1.000, write reduction 97.3% and RUL RMSE 42.7, 27.3 and 24.3 for the three predictors. Quality metrics are identical across seeds; only latency varies.
CMAPSS FD002 and FD003 (5 seeds each)	DSP, $d = 16$; FD002 standardized per operating condition (scaler fitted on the fit units only)	Health AUROC 0.994 (FD002) and 0.987 (FD003), against 0.999 on FD001; end-of-life retention 1.000 on both, as on FD001; write reduction 96.1% and 95.6%, against 96.7%	The health score transfers to other subsets and the memory keeps every unit's end-of-life state. RUL RMSE (memory lookup) is 53.4 and 41.9 against 32.1 and 22.3 for naive storage: with so few items, each spans the whole RUL range and the lookup collapses to a near-constant prediction, as on FD001. RUL from the health score is 36.8 and 29.8, and 34.5 and 25.6 with unit age. An earlier version of these runs reported retention 0.062 and 0.120 and write reduction 78.5% and 80.0%; the strict step-inflation estimate rested on 17 and 14 warm lag pairs and acted as white noise, so the step detector fired on slow within-unit motion (Section 3.7). The reservoir baseline retains 0.195 and 0.416 at the same 256-slot budget, and the DSP encoder is deterministic, so seeds vary only the reservoir baseline. Two cases that earlier raised "fewer than 10 difference pairs" now run with unit-grouped step inflation: FD002 without normalization (health AUROC 0.624, retention 1.000, RUL RMSE 39.6 from health and age) and FD004 with it (health AUROC 0.988, retention 1.000, write reduction 95.8%, RUL RMSE 38.9). All five are single configurations on consumed benchmarks.
MetroPT-3 (5 seeds)	CNN encoder, frozen alarm policy	4 of 4 events in every seed; 12.6 false alarms per week (6.9–18.3); mean first alarm minus report start –7.40 h (–16.6 to +8.6)	492 of 495 event windows fall in sustained load, which commissioning never contained, and 87% of healthy sustained-load windows already exceed the health setpoint. Within sustained load, clean windows against the 24 h before a report give frozen seed-0 DSP AUROC 0.07 (health) and 0.35 (novelty). At 12.6 false alarms per week, an alarm inside a 24 h look-back is expected by chance, so lead times are not evidence of early warning.
MetroPT-2	MetroPT-3 design replayed	0 of 5 events under the selected alarm design (January 0 of 3, April 0 of 2)	A less conservative alarm rule catches both April events at 17.8 false alarms per week, so the outcome depends on alarm design. The replay uses frozen seed-0 DSP scores.
SKAB (seed 0)	DSP, frozen alarm policy	5 of 34 fault experiments; AUROC 0.546; AP 0.419; 135 false alarms per week	All five detections come 28–114 s after fault onset. The weekly rate is extrapolated from 5 episodes over about 6.2 clean hours.
SCANIA Component X (seed 0)	Frozen alarm policy	22.8% recall on classes 1–4 at the final row; 19.3% alarm rate on class 0; AUROC 0.541	Coarse repair-proximity classes cannot measure lead time or early warning.

Dataset	Configuration	Primary result	Interpretation
EngineAD (10 seeds per width)	CNN encoder, $d \in \{8, 16, 32\}$, 30 epochs, 95th-percentile threshold	Held-out-truck pooled segment AUROC near chance: seed medians 0.539, 0.556, 0.537 for d = 8, 16, 32 ($d = 16$ range 0.540–0.567); median AP 0.176, 0.200, 0.179 against an anomaly base rate of 0.176	The released principal components were fitted on the whole fleet, including the held-out trucks, so preprocessing leaks across splits. Labels are per segment and 135 segments mix normal and anomalous points. Only five trucks are held out, and segment labels give no event onsets, so event recall and false alarms per week cannot be computed.

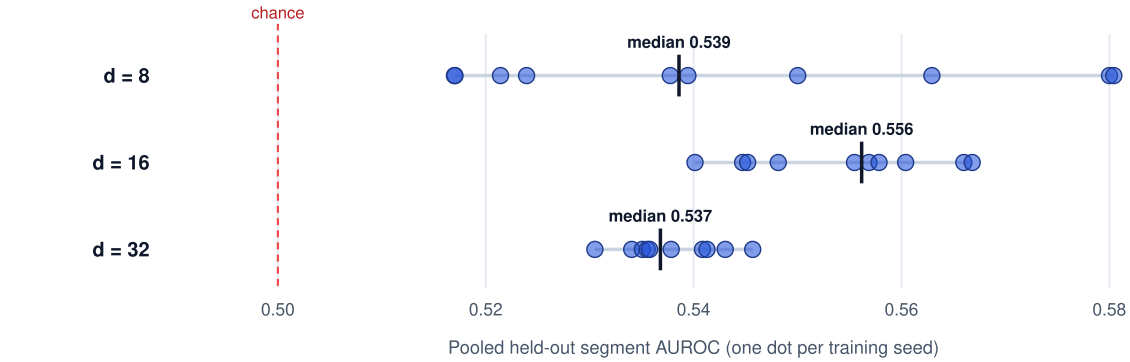


Figure 5. EngineAD pooled held-out segment AUROC for ten training seeds at each latent width; each dot is one seed, not one test vehicle.

5.4 SKAB Embedding Width

Table 6. SKAB, frozen alarm policy, seed 0, varying DSP width.

Width	Events detected	False-alarm episodes	AUROC	AP	Clean time in alarm
$d = 16$	5 / 34	5	0.546	0.419	1.3%
$d = 24$	5 / 34	6	0.551	0.419	2.7%
$d = 32$	5 / 34	7	0.575	0.415	6.1%
$d = 48$	9 / 34	4	0.558	0.441	18.6%

The DSP encoder is deterministic: seeds 0–9 at $d = 48$ give identical metrics, so this is one result, not a distribution.

Noise-whitening ablation. We replaced the learned noise rotation (Algorithm 1) with the identity, keeping the PCA projection. On the synthetic stream (10 seeds, $d = 16$) the effect is small: retention stays at 1.000, write reduction moves from 86.19 to 86.00% and AP from 0.9955 to 0.9918. On SKAB it is large: at $d = 16$ detected events fall from 5 of 34 to 2 of 34, AUROC from 0.546 to 0.532 and AP from 0.419 to 0.383 (seed 0, frozen alarm policy). Whitening therefore matters on this real testbed and barely on the synthetic generator; we did not measure either dataset's noise correlation, so why is untested.

5.5 Alarm-Threshold Tuning and Missing State Coverage

A parameter sweep of 1,302 calibratable alarm-policy and corroboration-rule combinations on MetroPT-3 found exactly one setting meeting all three operational targets: 4 of 4 events detected, at most 2% clean alarm time and at most 1 false alarm per week. That setting uses a novelty threshold ten times looser than the frozen policy ($\alpha = 0.03$ against 0.003), gated by a calibrated-error corroboration vote with a 6-hour lookback (clean

alarm time 1.21%, 0.58 false alarms per week). The vote is true for 94.8% of clean exposure time, so it rarely removes anything. Excluding this row, the next-best setting misses the clean-time target, at 2.08% and 0.64 false alarms per week. With over 1,300 untested-in-advance combinations swept, one setting clearing a budget by a narrow margin through an almost-always-true gate is consistent with chance, not a demonstrated alarm-design improvement.

An earlier version of this sweep treated samples with no calibration baseline as normal rather than unknown. Under that two-valued handling, the frozen policy ($\alpha = 0.003$) with the same vote and a 6-hour lookback also met all three targets, at 1.06% clean alarm time and 0.95 false alarms per week. Auditing showed that the vote never fires in the sustained-load state (0 of 1,832 test windows there, against 4,598 of 9,219 and 6,417 of 10,505 in the other two states), because that state has no calibration baseline. The gate therefore acted as a recency cap on the other states, not as corroborated detection. Treating unjudged samples as unknown instead removes the frozen policy's qualifying result: its best 4-of-4 setting then has 4.62% clean alarm time. The looser-threshold setting above qualifies under both treatments. A gate that never activates in the state that matters can still improve a headline number.

5.6 Representation Depth: Proxy Screen Versus Full Pipeline

Recent work suggests that intermediate layers preserve localized anomaly signals that final bottlenecks smooth over [13]–[16]. To test this without test-label leakage, we added a multi-level CNN readout and first ran a label-free synthetic screen across 5 neural seeds (384 training windows, 96 selection windows injected with five fault families: level shift, drift, impulse, oscillation, freeze). On that screen the middle-layer readout ($d = 32$, 30 epochs) ranked first, with a median worst-family displacement score of **0.610**, against **0.373** for the multi-level readout and **0.356** for the final bottleneck. Training for 90 epochs degraded every configuration; over-specialization on waveform reconstruction is a plausible cause, but we did not test it.

Running the same five seeds through the full calibrated alarm pipeline overturned the screen (Table 7). Both datasets had already been used for development, so this is not independent confirmation, and the selection was exploratory with all variants retained.

Table 7. Intermediate CNN readouts through the full calibrated alarm pipeline (5 seeds, $d = 32$, 30 epochs). Cells are medians with the range over seeds in parentheses.

Dataset	Readout	Events detected	AUROC	AP	False alarms / week	Clean alarm share
SKAB	Bottleneck	**23.0** (18–34)	**0.623** (0.480–0.675)	**0.511** (0.441–0.585)	189.25	0.480
SKAB	Middle layer	20.0 (18–29)	0.570 (0.540–0.660)	0.505 (0.448–0.538)	243.32	0.440
SKAB	Multi-level	20.0 (13–33)	0.571 (0.553–0.687)	0.504 (0.467–0.518)	243.32	**0.310**
MetroPT-3	Bottleneck	4.0 (3–4)	0.989 (0.987–0.996)	0.676 (0.485–0.959)	0.69	0.022
MetroPT-3	Middle layer	3.0 (3–4)	0.990 (0.987–0.992)	0.617 (0.517–0.671)	**0.58**	**0.019**
MetroPT-3	Multi-level	**4.0** (4–4)	**0.992** (0.988–0.996)	**0.773** (0.541–0.956)	**0.58**	0.025

The middle layer had the lower event median on both datasets (SKAB 20 against 23; MetroPT-3 3 against 4), though on MetroPT-3 its median AUROC was marginally higher (0.990 against 0.989) and its false-alarm rate lower (0.58 against 0.69 per week). Seed spread was wide on SKAB (13–34 events across readouts), and no readout dominated across datasets. The bottleneck therefore remains the default.

5.7 Cross-Dataset Distillation Results

We evaluated single- and dual-teacher distillation on five gates covering seven datasets. Higher foundation-model AUROC frequently masks severe operational alarm failures.

1. **MetroPT-3 against a synthetic context mismatch.** Distilling MOMENT-1-small improved MetroPT-3 test AP from 0.6225 to 0.7942 (AUROC 0.9756 to 0.9775), in one run on data already consumed in development. It degraded the synthetic four-family probe AUROC from 0.6872 (ordinary CNN) to 0.5035 (edge-padded) and 0.5362 (resampled with instance statistics). The statistics-augmented teacher also lowered MetroPT-3 AP, from 0.6225 to 0.5846.
2. **TSB-AD-U external holdout (six series).** Resampled MOMENT with statistics edged out the ordinary CNN on pooled AUROC (0.9157 against 0.9111), where long series dominate, but lost on equal-series AUROC (0.6316 against 0.6476), averaged over the six series (Figure 6).
3. **MFPT and CWRU bearing gates (ranking against calibration).** Combining MOMENT (514 dimensions) and Chronos-2 (32 dimensions) into a 46 KB student raised MFPT window AUROC from 0.7948 to 0.8253 (file AUROC 0.85 to 0.90) and reached 1.0000 on CWRU for all models. However, every model flagged the single held-out normal recording on both datasets (normal-file false-alarm rate 1.0), because the training-normal threshold treated that healthy recording as a fault. On CWRU the held-out recording is at 3 HP while training used 0–2 HP, which is consistent with an unseen-load explanation. No load is recorded for the MFPT files, so the MFPT cause is untested.
4. **CMAPSS run-to-failure gate (AUROC against alarm timing).** One sensor, engines 1–40 for training and 41–50 held out to failure, three consecutive anomalous windows required. Teacher-distilled students achieved higher window AUROC (0.9890–0.9902 against 0.9823 for the ordinary CNN). All four models raised a sustained alarm before failure on 10 of 10 engines. The RUL-20 horizon metric counts an engine only if a sustained alarm starts at RUL $\leq$ 20: the ordinary CNN scores 7 of 10, the Chronos student 3 of 10, and the MOMENT and two-teacher students 2 of 10. The students' alarms start earlier and more often (13–19 episodes before the horizon against 7 for the CNN, over 1,723 healthy cycles; median warning cycles before the horizon 9 for the CNN and the MOMENT student, 23.5 and 23 for the Chronos and two-teacher students). The gate cannot separate early warning from false alarm, so the 7-of-10 against 2-of-10 gap reflects alarm-onset timing, not lost lead time.
5. **SKAB single-channel against eight-channel fusion.** In an exploratory gate on one seed, single-channel runs (teacher-distilled and no-teacher CNN) detected 0 of 34 events with a persistent alarm starting inside the fault interval (AUROC 0.463–0.471, 5.47–8.75 false alarms per hour). Running all 8 channels through the multi-level CNN without teachers cut healthy false alarms by 58.3% relative to the 8-channel bottleneck (6.57 to 2.74 per hour) and raised AUROC to 0.5270, but strict in-interval detection stayed at 0 of 34. These rates are per hour of healthy calibration data. This criterion is stricter than the active-in-interval criterion behind Table 7, so the two are not comparable.

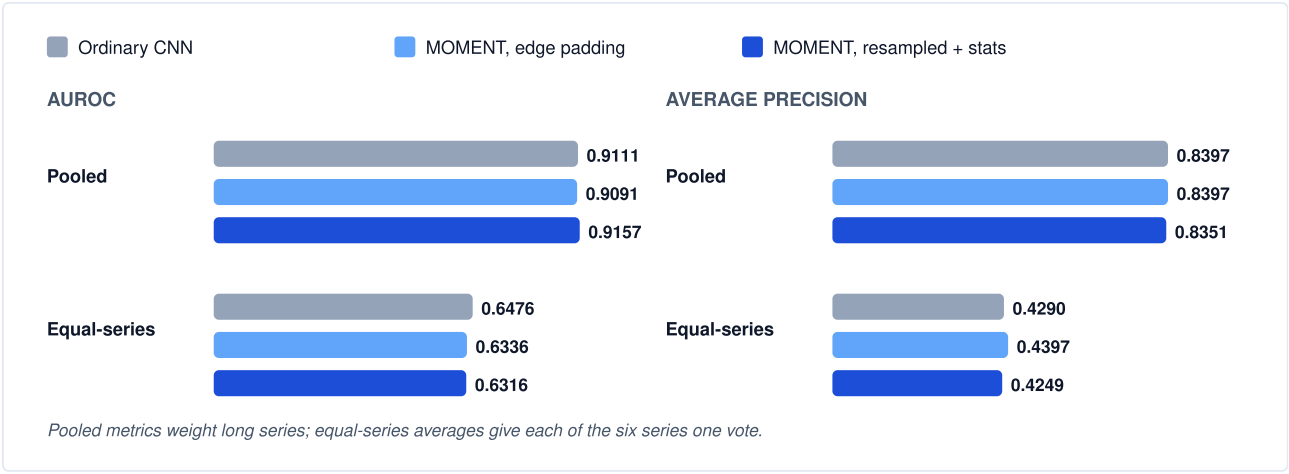


Figure 6. TSB-AD-U pooled and equal-series AUROC and AP for six frozen series.

5.8 Role Separation on SKAB

On the 8-channel SKAB split, where all three CNN representations scored 0 of 34 strict detections, the role-separated architecture detected **10 of 34** events at 6.19 false alarms per hour (at-least-two policy) and **8 of 34** at **1.59** false alarms per hour (dynamic-and-health policy). The comparison is not like-for-like: the CNN false-alarm rates are per hour of healthy calibration data, the role figures per hour of clean test data, and the CNN gate is a single seed.

5.9 Domain Specialists on OS-Metrics

To test role separation on multi-subsystem operational telemetry we used OS-Metrics (40,320 samples at 15 s over 7 days, 42 annotated disturbances).

Numerical diagnostics. A six-channel ingestion first crashed with a singular-matrix error during noise whitening. The pre-fit diagnostic of Section 3.7 flagged the six-channel matrix as *singular* (rank 5 of 6, condition number 7.15×10^{15}), because one network response-time channel was an exact linear combination of the other two, and certified the five primitive channels as *moderately conditioned* (rank 5 of 5, condition number 4.76×10^{8}). The diagnostics were computed on all rows, not only the fit split, so they describe the dataset rather than a strictly pre-fit measurement.

Even on the five primitive channels, pooling all metrics into one global role encoder underperformed the DSP baseline (24 of 42 recall at 42.0 false alarms per week against 27 of 42 at 26.6), plausibly because localized subsystem spikes were diluted by quiet channels (untested). Partitioning into CPU, memory and network specialists improved on both (Table 8).

Table 8. OS-Metrics chronological split: DSP baseline, all-channel role encoder and domain specialists. Recall is over all 42 annotated events.

Architecture	Strict event recall	2-hour recall	False alarms / week	Window AUROC	Root-cause attribution	Latency / window
DSP ($d = 16$) baseline	27/42 (64.3%)	30/42 (71.4%)	26.6	0.8845	n/a	n/a
All-channel role encoder	24/42 (57.1%)	30/42 (71.4%)	42.0	0.8731	n/a	6.81 µs
Domain specialists	**29/42 (69.0%)**	29/42 (69.0%)	**19.6**	**0.8919**	30/30 contain the true subsystem; 14/30 unique	6.27 µs

Domain specialists achieved the highest strict recall and the lowest false-alarm rate (Figure 7 compares the three architectures). Their 6.27 µs per-window batch scoring time (excluding fusion) compares with 6.81 µs for the all-channel role encoder; the DSP latency was not measured. Window AUROC is highest for the specialists (0.8919), but window AP is not: DSP reaches 0.8577, the all-channel role encoder 0.7986 and the specialists 0.7847.

Three limits apply to these figures. First, 12 of the 42 annotated disturbances begin before the test split, in the fit and calibration days, where nothing is scored, so every method scores 0 of 12 on them by construction. On the 30 test-split events the strict recall is 27/30 for DSP, 24/30 for the all-channel role encoder and 29/30 for the specialists. Second, windows overlapping annotated events were removed from the fit and calibration sets using the event labels. Third, "attribution" counts an event when any specialist raises a single-window flag, with no persistence requirement, and counts it correct when the true subsystem is among the triggers. By that measure 30 of 30 test-split events are attributed correctly, but only 14 have a single trigger; 13 have two and 3 have all three, and the CPU specialist fires on 13 of the 23 non-CPU events. The fused strict-recall figure of 29/30 requires persistence, so one test-split event is flagged without being detected under that stricter rule.

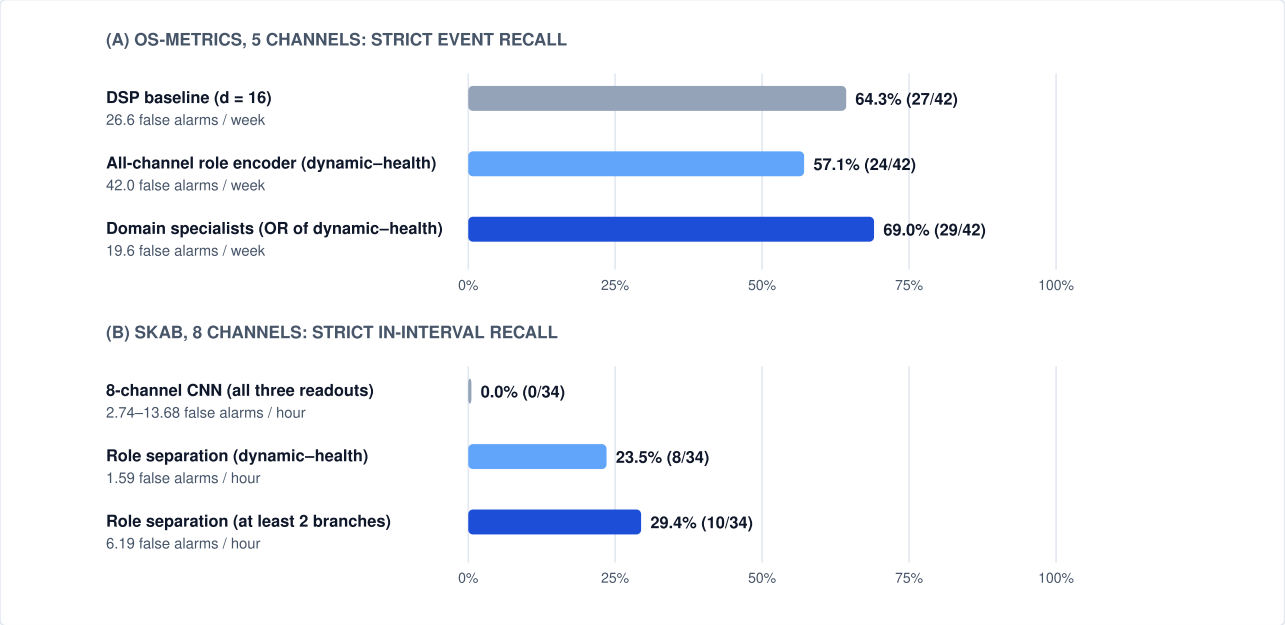


Figure 7. Strict event recall of domain specialists and role separation against monolithic encoders on OS-Metrics (A) and SKAB (B). Panel A uses all 42 annotated events as the denominator, of which 30 lie in the test split. Panel B's CNN row summarizes the three readouts of Section 5.7; it is not comparable to Table 7.

Per-subsystem results show where each detection comes from; the event counts sum to the 42 disturbances (13 + 12 + 17).

Table 9. Per-specialist results on OS-Metrics.

Specialist	Assigned channels	Strict event recall	False-alarm episodes	Window AUROC
CPU	CPU usage	7/13 (53.8%)	6	0.9589
Memory	Memory usage, memory used	8/12 (66.7%)	8	0.9641
Network	Two network response-time channels	14/17 (82.4%)	12	0.8924
Fused (OR of specialists)	All five primitive channels	29/42 (69.0%)	14 (19.6/week)	0.8919

In isolated incidents, only the responsible specialist fired (for example, only the memory specialist for one memory event). In cascading disruptions, several specialists fired together, and in one such event all three did. Because the CPU specialist fires on 20 of the 30 test-split events, "cascading" is our reading of the trigger pattern, not an independently labeled event type. The per-specialist false-alarm counts in Table 9 are each measured against that specialist's own subsystem events and are not comparable to the fused 14.

5.10 Comparison Against External Streaming Baselines

Sections 5.1–5.9 compare µEngram against internal controls (naive, FIFO, reservoir, centroid clustering). Table 10 adds two external streaming-anomaly-detection baselines, half-space trees and MemStream [6], on MetroPT-3 and SKAB, under the DSP encoder and each method's own alarm configuration selected on MetroPT-3 development data, calibrated the same way as µEngram (Section 3.4).

Table 10. MetroPT-3 and SKAB against half-space trees and MemStream. DSP encoder; MetroPT-3 over 5 seeds, SKAB over 5 seeds for half-space trees and MemStream (µEngram's DSP score is deterministic, so 1 seed).

Dataset	Method	Events detected	False alarms / week	AUROC
MetroPT-3 (4 events)	µEngram	3 / 4	0.53	0.985
MetroPT-3 (4 events)	Half-space trees	1.6 / 4 (1–3)	0.46 (0.26–0.69)	0.889
MetroPT-3 (4 events)	MemStream	2.0 / 4 (1–3)	0.26 (0.21–0.32)	0.920
SKAB (34 events)	µEngram	5 / 34	135.2	0.546
SKAB (34 events)	Half-space trees	3.6 / 34 (2–5)	10.8 (0.0–27.0)	0.519
SKAB (34 events)	MemStream	1.0 / 34	0.0	0.545

Under this frozen, DSP-encoded configuration (distinct from the CNN-encoded configuration in Table 5), µEngram detects the most events on both datasets, at the cost of the highest false-alarm rate: 135.2 per week on SKAB against 10.8 for half-space trees and 0.0 for MemStream, and 0.53 per week on MetroPT-3 against 0.46 and 0.26. None of the three methods reaches both high recall and a low false-alarm rate on SKAB. All three alarm configurations were selected on MetroPT-3 development data, so this is a same-protocol comparison, not independent confirmation of any method.

5.11 DSP Feature-Group Ablation

The DSP encoder (Section 3.2) builds six features per channel: the mean, the log standard deviation, and four log band energies. To see which group carries signal, we dropped one group at a time and left everything else unchanged (robust scaling, PCA to $d = 16$, noise whitening, alarm policy). The full configuration reproduces the synthetic, SKAB and CMAPSS figures reported above, so the patch itself is neutral.

Table 11. Feature-group ablation, µEngram, DSP $d = 16$ requested, 256 slots. Synthetic: mean of 10 seeds. SKAB: seed 0, frozen alarm policy. CMAPSS: one run (the encoder is deterministic). Without band energies the synthetic stream has 12 features, so d is clamped from 16 to 12.

Features kept	Synthetic recall / AP	SKAB events (false-alarm episodes)	CMAPSS FD001 health AUROC / EOL retention	CMAPSS FD003 health AUROC / EOL retention
All six	1.00 / 0.9955	5 / 34 (5)	0.999 / 1.00	0.987 / 1.00
Without mean	1.00 / 0.9954	0 / 34 (0)	0.748 / 1.00	0.859 / 1.00
Without log std	1.00 / 0.9909	5 / 34 (6)	1.000 / 1.00	0.984 / 0.66
Without band energies	0.79 / 0.476	9 / 34 (9)	0.999 / 1.00	0.990 / 0.98

Table 11 shows that the groups matter differently by dataset. The mean is the carrier on SKAB and CMAPSS: removing it takes SKAB from 5 detections to none, FD001 health AUROC from 0.999 to 0.748 and FD003 health AUROC from 0.987 to 0.859, yet it costs nothing on the synthetic stream (AP 0.9955 to 0.9954), so the synthetic stream does not separate encoders that keep the mean from those that discard it. Removing the log standard deviation changes little on the synthetic stream, SKAB and FD001, but on FD003 it lowers end-of-life retention from 1.00 to 0.66 and write reduction from 95.6% to 90.6%, so FD003 retention depends on which features enter the distance. An earlier version of this table, produced before the step-inflation estimate required 100 warm lag pairs, showed the opposite sign (FD003 retention 0.12 with all features and 0.66 without the log standard deviation) and several "fit fails" cells; those were artifacts of the strict estimators (Section 3.7).

Two cautions apply. Dropping the band energies leaves two features per channel, which is 12 features on the synthetic stream, 16 on SKAB and 28 on FD001. Twelve cannot support $d = 16$, so the synthetic encoder now runs at $d = 12$ (it earlier failed) and gives recall 0.79 and AP 0.476; that is a lower-dimensional encoder as well as a different feature set. At 16 the PCA does no reduction on SKAB, and the 9 of 34 detections (against 5) come with 9 false-alarm episodes (against 5), so this is a different operating point of one seed rather than an improvement; the change is also confounded with the effective dimension. The SKAB and CMAPSS columns are single runs, and the suggested mechanism (that the level feature carries slow drift) is untested.

5.12 Confirmation on 28 Server Machines

To test whether the findings above depend on the few datasets used to develop them, we ran the frozen alarm policies on the Server Machine Dataset (SMD) [30]: 28 server machines with 38 resource-usage channels each, a normal-only training file and a labelled test file per machine. SMD was not used to choose any parameter. The alarm configurations of µEngram, half-space trees and MemStream are those selected on MetroPT-3 development data (Section 5.10), so this is a same-protocol check on unseen assets, not tuned operation. The files carry no timestamps, so a one-minute grid is assumed (train, then test), and false-alarm rates per week are per 168 hours of that grid.

Table 12. SMD, 28 machines, 327 labelled events, DSP encoder, 256 slots. Events and false alarms are pooled over machines; AUROC is the unweighted mean over machines. Half-space trees and MemStream: mean (range) over 5 seeds. The last three rows drop one DSP feature group (Table 11).

Method	Events detected (of 327)	False alarms / week	Mean AUROC
µEngram, $d = 16$	35	1.23	0.832
µEngram, $d = 36$	41	1.23	0.832
Half-space trees	35.0 (30–41)	0.55 (0.45–0.61)	0.767
MemStream	9.0 (7–11)	0.05 (0.03–0.08)	0.764
µEngram without the mean	20	0.76	0.743
µEngram without log std	38	0.76	0.839
µEngram without band energies	39	0.78	0.845

Table 12 shows three results. First, event-level detection is weak for every method: µEngram detects 35 of 327 events (11%), at least one event on 17 of 28 machines and all events on 2. It matches half-space trees on events at about twice the false-alarm rate, and ranks windows better (mean AUROC 0.832 against 0.767 and 0.764). Second, the width regression of Section 5.2 does not reproduce: $d = 36$ detects 41 events against 35 at the same pooled AUROC (paired by machine, 18 machines improve and 10 worsen, mean change +0.0001), so widening the projection is not harmful in general. Third, the mean carries signal here as on SKAB and CMAPSS FD001: dropping it costs 15 events and 0.089 AUROC, while dropping log std or band energies does not hurt (38 and 39 events, AUROC 0.839 and 0.845).

SMD also allows a per-asset test of whether a label-free descriptor predicts which machines depend on a feature group. The hypothesis, fixed before the run, was that dropping a group costs more AUROC on machines where that group has higher signal-to-noise. The descriptor is computed from fit windows only: per channel, the robust variance of the feature across windows divided by the noise variance from successive-window differences, then the median over non-constant channels of the base-10 logarithm. The effect is AUROC with all features minus AUROC with the group dropped. Across the 28 machines the rank correlations are -0.16 for the mean (permutation $p = 0.40$), $+0.38$ for log std ($p = 0.047$) and $+0.34$ for band energies ($p = 0.077$; 10,000 permutations). With three tests the threshold is $0.05/3 = 0.017$, so none is significant and the hypothesis is not supported: this descriptor does not tell us which machines need which features.

5.13 Confirmed Latent Adaptation: Gating Is Not Optional

Section 6 proposes keyed state superseding: an active baseline updates only from windows an operator confirmed normal. We implemented it (`confirmed_adaptation.ConfirmedLatentAdapter`, wired into the live ingest path) and tested whether the gate matters, on a synthetic 8-dimensional latent stream built for this question: 1,500 commissioning windows, 1,000 calibration windows, then a 6,000-window test stream per seed (10 seeds) with a benign linear drift (4σ over the stream, 25% of healthy windows marked confirmed) and five independently-directed wear ramps (reaching 8σ, never confirmed). An *ungated* control marks every window confirmed, ramps included, to show what happens when confirmation is not a real gate.

Table 13. Confirmed latent adaptation, 10 seeds, mean over seeds. Health alarms use a conformal threshold at $\alpha = 0.003$ from the calibration split.

Arm	Scenario	Health false-alarm rate	Health recall on ramps	Health AUROC	Ramps with ≥1 alarm (of 5)
Frozen (no adapter)	drift	9.7%	87.2%	0.959	5.0
Gated	drift	0.5%	86.6%	0.993	5.0
Ungated	drift	2.9%	1.7%	0.546	3.4
Frozen (no adapter)	no drift	0.3%	87.4%	0.994	5.0
Gated	no drift	0.4%	87.0%	0.993	5.0
Ungated	no drift	3.2%	1.0%	0.527	3.9

With benign drift, the frozen reference pays for never adapting: a 9.7% false-alarm rate on healthy windows, because "healthy but drifted" keeps tripping a threshold calibrated before the drift. Gating removes nearly all of it (0.5%) at no cost to ramp recall (86.6% against 87.2%): the confirmed-normal windows track the drift while the unconfirmed ramps are still flagged. The ungated control is the failure this gate exists to prevent: with every window accepted, the adapter partly absorbs the ramps themselves into the baseline, and recall collapses to 1.7% with AUROC near chance (0.546). Without drift, gating costs nothing (false-alarm rate and recall both match the frozen arm within noise), so the gate is not merely cautious, it is doing real work exactly where drift is present and failing exactly as designed when confirmation is not trustworthy.

5.14 Restart Recovery: Detector State Is Part of the Memory

Sections 5.1–5.13 assume an uninterrupted process. We tested what a restart does, using the first 64 test windows of one SMD machine (`machine-1-1`, 38 features per window, window means as inputs, no labels read) with a restart injected at window 32. Three arms received identical inputs and a Lite-like idempotent event store: **Lite-only** (events delivered directly), **sidecar** (a durable outbox and exact vector file, but no detector state) and the **capsule** (Section 3.5). The reference is an uninterrupted run, which emits three lifecycle records: a new candidate at step 0, a promotion at step 4 and an alarm-open at step 39. We compare engine arrays

exactly and compare lifecycle records as sets of idempotency keys, not as counts. Each of three restart points (before the engine consumed the logged window, after it consumed the window but before a checkpoint, and after a checkpoint) was run five times on each of four hosts (x86 desktops and laptops: i7-6700, i3-6100, i3-3110M and i3-7100), giving 60 runs per arm; Table 14 reports them.

Table 14. Restart recovery, 60 runs per arm (4 hosts × 3 restart points × 5 repeats). Every host and restart point gave the same outcome.

Arm	Engine state equals uninterrupted	Lifecycle keys equal reference	Missing records	Extra records
Lite-only	0 / 60	0 / 60	1	2
Sidecar	0 / 60	0 / 60	1	2
Capsule	60 / 60	60 / 60	0	0

Without detector state the restarted engine treated the machine as new: it re-emitted the new-candidate record (step 32) and the promotion record (step 36), and it never emitted the alarm-open record that the uninterrupted run raised at step 39. A count comparison would have hidden this: the Lite-only arm held four records against the reference's three, which is two spurious records and one missing. The capsule matched the reference in all 60 runs, with identical engine arrays and the same three keys. We do not compare retrieval between arms: the stand-in store returned only its newest record on recall, whereas the real store returned all four.

Why the alarm is lost. The reference's score climbs from 0.17 at step 32 to 1.21 at step 39 as the stream drifts away from the baseline learned on steps 0–31. The restarted engine has no baseline: its scores are undefined for five steps and then stay below 0.25, because it learns the already-drifted windows as normal. Replaying pre-restart windows into a fresh engine gives a dose-response. With 8 or fewer windows no alarm opens; with 16 the alarm opens late (step 44) and flaps, with two open and two clear records; with 24 or 32 the records after the restart equal the reference's exactly. This supports, without proving, the explanation that a cold restart re-baselines on whatever it sees first, so a drift that began before the restart is absorbed as normal (one machine, one restart position).

Generality. To check that this is not an artifact of one machine or one restart position, we repeated the comparison on all 28 SMD machines with a 256-window prefix and 16 evenly spaced restart positions, giving 448 machine-position cells. The capsule matched the uninterrupted run, in engine state and lifecycle records, in 448 of 448 cells. Lite-only and sidecar matched in none and lost at least one alarm-open record in 406 of the 448 cells (90.6%); on average 3.6 of the reference's 8.2 alarm-open records were missing after a restart, and every cell held spurious records. Restarts at windows 16 through 210 lost an alarm-open on 93 to 100% of machines, falling to 82% and 39% for the last two positions, where little remains to alarm on. Against the real EdgeMemory Lite library on three machines (24 cells) the capsule again matched in 24 of 24 and the stateless arms lost an alarm-open in 18 of 24. The alarm records here come from a fixed threshold, not the calibrated policy of Sections 5.3–5.5, so this measures divergence from an uninterrupted run, not missed faults.

Process death. The restarts above close and reopen the store inside one process. To test real process death we SIGKILLed a capsule-backed ingest loop at a random moment, started a second process that recovered, drained the outbox and finished the stream from the recovered cursor, and compared with an uninterrupted run (Table 15). Each trial checks four things: lifecycle records equal the reference, the vendor holds no duplicate rows, engine arrays equal the uninterrupted run after recovering from the final on-disk state, and SQLite reports no corruption.

Table 15. SIGKILL recovery campaigns on one host with temporary storage on ext4, three SMD machines, 128-window prefix. A trial counts as killed mid-stream if the loop had not finished when the kill landed. Every trial in every campaign passed all four

checks.

Campaign	Trials	Killed mid-stream	Input logged, not checkpointed	Vendor ahead of checkpoint	Passed all checks
Stand-in store, natural timing	150	127	86	14	150
Stand-in store, 40 ms pause after each vendor commit	150	132	96	47	150
Real EdgeMemory Lite, natural timing	150	123	73	9	150
Real EdgeMemory Lite, 40 ms pause after each commit	150	140	101	46	150
Normal sync, checkpoint every window (stand-in, pause)	60	52	42	29	60
Full sync, checkpoint every 16 windows (stand-in, pause)	60	47	45	34	60
Normal sync, checkpoint every 16 windows (stand-in, pause)	60	47	43	39	60

Natural timing rarely killed the process between a vendor commit and the matching checkpoint (14 and 9 trials), which is the most dangerous boundary, so we injected a 40 ms pause after each vendor commit; that raised the count to between 29 and 47 trials per campaign. The real vendor store recovered from every kill without error. SIGKILL is not power loss: it leaves the operating system's page cache intact, so torn writes and lost unsynchronized data remain untested, and the relaxed rows show only that those configurations survive process death. **Modeled power loss.** SIGKILL leaves the operating system's page cache intact, so we added a crash-image campaign. An LD_PRELOAD shim recorded the capsule's file operations, and the model replayed those operations into images cut at whole-syscall write boundaries. It does not model sub-write tearing, block-device reordering below the page cache, or flash behavior. The campaign used three SMD machines, 48-window prefixes at checkpoint intervals 1 and 16, 400-window prefixes at interval 1, both synchronization modes, and two negative controls. SMD labels were never read; results are shown in Table 16.

Table 16. Modeled crash-image recovery. The consistency columns count crash images; `Max lost` counts source windows.

Configuration	Trials	Capsule consistent	Images losing acknowledged windows	Max lost	Final vendor consistent
`FULL`, every window, 48-window prefix	1,491	1,491 / 1,491	0	0	1,491 / 1,491
`FULL`, every 16, 48-window prefix	1,471	1,471 / 1,471	0	0	1,471 / 1,471
`FULL`, every window, 400-window prefix	337	337 / 337	0	0	337 / 337
`NORMAL`, every window, 48-window prefix	1,491	1,491 / 1,491	949	47	1,491 / 1,491
`NORMAL`, every 16, 48-window prefix	1,472	1,472 / 1,472	766	48	1,472 / 1,472
`NORMAL`, every window, 400-window prefix	337	337 / 337	223	169	337 / 337
`sync-off` negative control	481	481 / 481	337	48	481 / 481
`no-journal` negative control	483	317 / 483	199	48	317 / 483

All `FULL` and `NORMAL` images remained internally consistent in this model. `NORMAL` nevertheless lost acknowledged input windows in 1,938 images; `FULL` lost none. The no-journal control produced 166 inconsistent images, including malformed SQLite databases. These results establish a modeled whole-syscall recovery property and make the `NORMAL` durability risk concrete; they do not establish hardware power-loss safety.

Cost on real storage. An earlier version of this section attributed a five-to-six-fold restart cost to durable commits. That measurement used temporary storage on tmpfs, where fsync is nearly free, and reflected CPU work in checkpoint serialization; on tmpfs full and normal synchronization cost the same (Table 17). On disks the cost is very different. Full synchronization costs 7.8 and 9.9 ms per window on the two SSD hosts, 105 ms on a hard disk and 490 ms on a slow overlay filesystem, against 0.14 to 0.44 ms for the memory-only pipeline. In the restart experiment on the same disks, completing the second half of the 64-window stream took 208 ms and 379 ms on the two SSDs, 3.4 s on the hard disk and 2.3 s on the overlay filesystem, against 7 to 22 ms for the Lite-only arm; the overlay host's storage was about six times faster during that run than during the cost study, which is why its restart time is below the hard disk's. A longer checkpoint interval removes the second commit per window but not the first, because the input log commits every window, so full synchronization cannot fall below about one fsync per window: 56 ms on the hard disk even with a 64-window interval. Normal synchronization avoids this at 0.3 to 2.9 ms per window on every disk, but it gives up durability of the most recent commits on power loss, which we could not test; the process-death campaigns above show only that it is safe against SIGKILL. A longer interval also lengthens the worst-case replay, up to the interval minus one windows: on the hard disk under full synchronization a 64-window interval took 438 ms to recover against 26 ms with a checkpoint every window.

Table 17. Mean ingest cost per window (ms) of the capsule by synchronization mode and checkpoint interval, against the memory-only pipeline. Medians over interleaved repeats: 5 repeats of 256 windows on the first SSD host, 3 repeats of 64 windows on the other three. The hard-disk host is a rotational disk; the overlay host's storage varied roughly sixfold between runs, so its figures are indicative only.

Host (temporary storage)	Memory-only	Full, every window	Full, every 16	Full, every 64	Normal, every window	Normal, every 16
msi (ext4, SSD)	0.14	7.77	3.64	3.23	1.47	0.31
lenovo (ext4, SSD)	0.38	9.95	4.10	3.61	2.93	0.75
asus (ext4, hard disk)	0.44	104.61	56.99	56.42	1.76	0.73
hp (overlay)	0.14	489.87	243.93	225.14	1.52	0.35
msi (tmpfs)	0.48	2.94	1.03	0.82	2.80	0.66
lenovo (tmpfs)	0.38	2.76	0.73	0.62	2.75	0.72
asus (tmpfs)	0.14	1.36	0.34	0.28	1.36	0.34
hp (tmpfs)	0.13	1.27	0.32	0.27	1.27	0.32

What this does not show. The correctness results rest on SMD server metrics only, and the process-death, real-store and crash-image campaigns do not establish embedded behavior. The alarm records come from a fixed threshold, so the operational value of an alarm-open record was not evaluated, and this is a systems-recovery result that says nothing about detection accuracy. The crash-image model operates at whole-syscall write boundaries; sub-write tearing, block-device reordering, flash behavior and real hardware power cuts remain untested. The hosts are x86 desktops and laptops, not embedded targets, so the times say nothing about embedded latency, memory or flash endurance. One further host was unreachable and is not covered. The first four-host timing runs used tmpfs and did not record the storage; the on-disk runs cover the same four hosts.

6. DISCUSSION AND LIMITATIONS

What the evidence supports. The bounded slot memory does what it is designed to do under the synthetic generator's assumptions: it keeps rare episodes, suppresses routine writes, and needs candidate promotion to do so. On public data, the evidence is about the gap between ranking and warning. Strong AUROC coexists with healthy recordings flagged as faults (bearings), no pre-report signal inside an uncommissioned state (MetroPT-3), and alarms that start earlier than a maintenance horizon (CMAPSS). A proxy screen that looked principled (the label-free intermediate-layer screen) was overturned by the full pipeline, and a gate that never activated improved a headline number (Section 5.5).

Threats to validity.

- *Single workloads.* Most field results are single seeds on single assets or testbeds. OS-Metrics is one workload with annotated disturbances, not operator-adjudicated equipment faults. Several feature and policy changes that helped one dataset regressed another.
- *Development reuse.* The alarm policy was selected on MetroPT-3 and then applied unchanged elsewhere, so MetroPT-3 results are development results. The CNN readout study reuses datasets already used for development.
- *Labels in fitting.* OS-Metrics removes fit and calibration windows that overlap annotated events using the event labels.
- *EngineAD.* The study is segment-level and exploratory: the released features leak preprocessing across splits, labels carry no event timing, and only five trucks are held out.
- *Modeled writes.* The write reduction counts operations in a benchmark model. Hardware flash endurance and write amplification were not measured.
- *Metric sensitivity.* Event-level outcomes depend on the detection criterion (strict against active-in-interval) and on the alarm design, as the MetroPT-2 and CMAPSS results show.
- *Estimator support.* An estimate resting on few pairs can silently change a headline result: the strict step-inflation estimate on FD001 $d = 24$, FD002 and FD003 moved end-of-life retention between 0.06 and 1.00 (Section 5.3). The 100-pair minimum follows from the variance of the estimate, not from these results, and the CMAPSS cells are single deterministic configurations on consumed benchmarks. Retention rose while the memory-lookup RUL worsened, so the two goals trade off.
- *A retracted early-warning claim.* An earlier internal recap reported 63.8 hours of warning and zero false alarms over four healthy weeks on a public bearing run-to-failure set (NASA IMS). Inspection showed the supporting script hard-coded its own degradation indices, used the final sample as the failure reference and evaluated against those assumptions, and its 7-day run had three false-alarm episodes. We withdrew the claim and report nothing from that set: without independently adjudicated onset labels, a scripted changepoint measures a time trend, not detection before a physical failure.
- *Restart experiment scope.* Section 5.14 rests on SMD server metrics: one machine for the mechanism probe, 28 machines for the restart sweep and three for the process-death and real-store campaigns. Restarts are simulated by reopening or by SIGKILL, never by a power cut; the event store is a stand-in except in the real-library runs; and the hosts are x86 machines whose timings carry run-to-run spread of tens of percent, up to sixfold on one overlay filesystem.

Toward a sensor-state memory database. An earlier draft of this section proposed five primitives as unimplemented future work. Since then we implemented four of the five in the reference library and measured one of them; the identity keys default to 0, so none of this changes the public-benchmark results in Sections 5.1–5.12, all of which run a single identity per stream.

1. *Hybrid temporal-vector-severity retrieval:* implemented as a ranking function, $S(q, m) = w_s \,\mathrm{sim}(q, v_m) + w_t \, 2^{-\Delta t / h} + w_r \, \tanh(r_m)$, wired into the retrieval call, with asset, subsystem and mode as strict filters, never

score terms. Stored events carry a quantized int8 latent snapshot inside the existing 1,024-byte metadata budget, included only when it still fits. Tested on the scoring math and on end-to-end ranking against a real in-memory backend; not benchmarked for retrieval quality, because no relevance-labeled query set exists for this task.

2. *Identity-keyed compaction:* slots now carry an asset/subsystem/mode/schema key; absorption, Ward-cost and duplicate merges, and dormant supersession are restricted to slots sharing a key, reusing the existing exact pooled-moment merge ($\mu = (n_1\mu_1 + n_2\mu_2)/(n_1 + n_2)$). This is a narrower claim than first proposed: merges still fire on the existing statistical cost functions, not on an operator-confirmation gate; "verified" compaction, in that stronger sense, is not implemented.
3. *Keyed state superseding:* implemented as a bounded, state-keyed latent offset, wired into the live ingest path, gated on an operator-confirmed-normal flag, a closed alarm and an already-commissioned key. Measured directly (Section 5.13, Table 13): on a synthetic stream with benign drift, gating cuts the health false-alarm rate from 9.7% to 0.5% at no cost to fault recall; an ungated control collapses recall to 1.7%.
4. *Pinned cross-modal raw pointers:* implemented. A raw-window pin now carries a required host-supplied checksum and an optional expiration; a pin-expiration query reports ids past deadline for a host janitor to act on. Unit tested; not benchmarked.
5. *Quota-bounded eviction with protected quotas:* implemented. A reference-counted protect/unprotect call shields a slot from eviction and from losing its item id in a merge; the sensor adapter protects an item for as long as any raw window pins it. Unit tested, including that two protected slots are never merged with each other; not benchmarked.

Deployment gates. Before transitioning µEngram from a Python reference design to a safety-relevant embedded deployment, five gates remain:

1. *Preregistered multi-asset field cohort:* held-out physical assets with independently adjudicated fault-onset timestamps and complete operating-mode coverage (startup, idle, full load, post-repair), without preprocessing leakage across vehicles or machines.
2. *Adversarial poisoning and shadow baselines:* replay slow degradation ramps against a frozen shadow reference to verify that quarantine prevents gradual wear from updating semantic healthy centroids. The semantic route was not exercised in the OS-Metrics runs, which contain no operator-confirmed normal labels.
3. *Abstention as a first-class output:* emit unknown-operating-mode or insufficient-calibration-support outputs instead of forcing out-of-distribution loads into false alarms or healthy summaries.
4. *Transactional cross-store crash recovery:* partly addressed. A single-host SQLite capsule restored uninterrupted detector state and lifecycle records in all simulated restarts, all 448 restart-sweep cells and all 780 SIGKILL trials, including against the real EdgeMemory Lite library (Section 5.14). Still open: power-cut and torn-write testing, the durability of relaxed synchronization, and one atomic unit spanning the memory checkpoint, the event store and circular-buffer pin reference counts.
5. *Hardware storage profiling:* p99 latency, peak RAM, flash write amplification, and approximate-search recall under high-water-mark eviction on target embedded flash.

7. CONCLUSION

We studied a bounded, deterministic memory for sensor streams and a set of encoders around it. The memory keeps rare episodes and suppresses routine writes under controlled assumptions, but its behavior on public benchmarks is governed less by ranking quality than by alarm design, calibration coverage and the criterion used to count a detection. Foundation-model students and intermediate-layer readouts improved ranking in

places without improving operational alarms, and a role-separated, per-subsystem design helped on the one multi-subsystem workload we tested. The deployment gates in Section 6, above all a preregistered multi-asset field cohort and hardware storage profiling, remain open.

DATA AND CODE AVAILABILITY

All datasets are public and are cited in Table 2: CMAPSS FD001 [20] (FD002, FD003 and FD004 come from the same archive), MetroPT-3 [21], MetroPT-2 [22], [23], SKAB [24], SCANIA Component X [25], OS-Metrics [26], MFPT [27], CWRU [28], TSB-AD-U [17], EngineAD [29] and SMD [30]. The synthetic generator is part of the released code. The reference implementation, the experiment scripts, the committed result files and step-by-step reproduction instructions that map each table and figure of this paper to a command are in the project repository at https://github.com/justdoGIT/uEngram, released under the MIT license. The repository's reproduction guide lists the dataset layout expected by the loaders. The synthetic $d = 36$ results and the SKAB width sweep were regenerated from the released code for this paper. Every CMAPSS result file and the feature-group ablation were regenerated at the released commit after the step-inflation support rule (Section 3.7); the naive, FIFO, reservoir and label-free centroid rows of each file are identical to the earlier output, and only the µEngram rows moved. For Table 7, a fresh run of the released pipeline reproduced 146 of the 150 underlying per-seed values (five metrics, three readouts, five seeds, two datasets) exactly; the other four differ by less than 4×10^{-6}. The EngineAD study (30 runs) was produced by a released script on a cloud GPU (1,962 s) and its result file is committed. We reran the released script on a fresh Colab T4 GPU (1,680 s), after verifying all 25 data files against the same manifest (identical manifest hash, protocol and label audit, same Python, NumPy, pandas and torch versions). The seed medians of pooled segment AUROC were 0.540, 0.555 and 0.537 for d = 8, 16 and 32 against 0.539, 0.556 and 0.537 in the original run, the medians of AP 0.177, 0.200 and 0.179 against 0.176, 0.200 and 0.179, and every one of the 30 per-run AUROC values lay within 0.005 of its original (mean difference 0.001). GPU kernels are not bitwise deterministic, so exact equality is not expected; both result files are committed. The original file records the sweep label rather than a hash of the script, so its match to the released script rests on its fields, not on a checksum.

The restart study of Section 5.14 is reproducible from the repository: its per-run artifacts and aggregates are committed, and the reproduction guide lists the commands. Each artifact records the replay-input digest and host metadata (CPU model, memory, Python and NumPy versions and, in the later runs, the filesystem backing the temporary directory), and the aggregator rejects a set that mixes inputs. The four hosts shared one input digest despite running two Python and NumPy builds. The first four-host runs used tmpfs and did not record the storage. Timings were taken on interactive machines and are not a benchmark of embedded hardware.

REFERENCES

1. Chhikara, P., Khant, D., Aryan, S., Singh, T., and Yadav, D. (2025). “Mem0: Building Production-Ready AI Agents with Scalable Long-Term Memory.” arXiv:2504.19413.
2. dos Santos Oliveira, A., da Silva, F. S. T., Ferreira, D. L., et al. (2026). “Looking at anomaly detection in EdgeAI through the up-to-date lens of academic and market perspectives.” *Cluster Computing*, 29, 251. doi:10.1007/s10586-026-05954-9.
3. Willsky, A. S. (1976). “A Survey of Design Methods for Failure Detection in Dynamic Systems.” *Automatica*, 12(6), 601–611. [PDF].
4. Chow, E. Y., and Willsky, A. S. (1984). “Analytical Redundancy and the Design of Robust Failure Detection Systems.” *IEEE Trans. Autom. Control*, 29(7), 603–614. doi:10.1109/TAC.1984.1103593.
5. Green, A. A., Berman, M., Switzer, P., and Craig, M. D. (1988). “A Transformation for Ordering Multispectral Data in Terms of Image Quality with Implications for Noise Removal.” *IEEE Trans. Geosci. Remote Sens.*, 26(1), 65–74. doi:10.1109/36.3001.

6. Bhatia, S., Jain, A., Srivastava, S., Kawaguchi, K., and Hooi, B. (2022). “MemStream: Memory-Based Streaming Anomaly Detection.” *Proc. ACM Web Conf. 2022*. doi:10.1145/3485447.3512221 · arXiv:2106.03837.
7. Bang, J., Kim, H., Yoo, Y., Ha, J.-W., and Choi, J. (2021). “Rainbow Memory: Continual Learning With a Memory of Diverse Samples.” *CVPR 2021*, 8218–8227. [CVF].
8. Elenter, J., NaderiAlizadeh, N., Javidi, T., and Ribeiro, A. (2024). “Primal Dual Continual Learning: Balancing Stability and Plasticity through Adaptive Memory Allocation.” arXiv:2310.00154v2.
9. Aghasanli, A., Li, Y., and Angelov, P. (2025). “Prototype-Based Continual Learning with Label-free Replay Buffer and Cluster Preservation Loss.” arXiv:2504.07240.
10. Kim, M., Bhaumik, K. K., Ali, A. A., and Woo, S. S. (2024). “MIXAD: Memory-Induced Explainable Time Series Anomaly Detection.” arXiv:2410.22735.
11. Goswami, M., et al. (2024). “MOMENT: A Family of Open Time-series Foundation Models.” *ICML 2024*. arXiv:2402.03885 · [Code].
12. Ansari, A. F., et al. (2025). “Chronos-2: From Univariate to Universal Forecasting.” arXiv:2510.15821 · [Model Card].
13. Han, C. S., and Lee, K. M. (2025). “Leveraging Intermediate Representations of Time Series Foundation Models for Anomaly Detection.” arXiv:2509.12650.
14. Yang, L., Lu, L., Liu, C., et al. (2025). “Interactive exploration of CNN interpretability via coalitional game theory.” *Scientific Reports*, 15, 9261. doi:10.1038/s41598-025-94052-8.
15. Park, J., and Kang, S. (2026). “PaAno: Patch-Based Representation Learning for Time-Series Anomaly Detection.” arXiv:2602.01359v2.
16. Yoon, S., Kim, J., Ha, J., and Ko, Y. M. (2026). “Patch-based Memory Gate Model in Time Series Foundation Model.” arXiv:2509.18751v4.
17. TSB-AD Project. “A Comprehensive Benchmark for Time-Series Anomaly Detection.” https://thedatumorg.github.io/TSB-AD/. The univariate collection TSB-AD-U is distributed at https://www.thedatum.org/datasets/TSB-AD-U.zip.
18. Si, H., Li, J., Pei, C., et al. (2024). “TimeSeriesBench: An Industrial-Grade Benchmark for Time Series Anomaly Detection Models.” arXiv:2402.10802v3.
19. Kim, J., Yoon, S., Ko, Y. M., Kim, Y., Kim, S. I., and Tae, J. (2026). “Complexity- and Statistics-Guided Anomaly Detection in Time Series Foundation Models.” *ICLR 2026*. [Paper].
20. NASA. “Turbofan Engine Degradation Simulation Data Set (CMAPSS FD001).” NASA Open Data Portal. https://data.nasa.gov/dataset/cmapss-jet-engine-simulated-data.
21. MetroPT-3 dataset (Metro do Porto air compressor, 2020). UCI Machine Learning Repository. https://archive.ics.uci.edu/dataset/791/metropt+3+dataset.
22. Veloso, B., et al. (2022). MetroPT dataset for predictive maintenance. *Scientific Data*, 9, 764. Data: doi:10.5281/zenodo.7766691.
23. Jakobs, M., Veloso, B., and Gama, J. (2025). arXiv:2502.07394.
24. SKAB: Skoltech Anomaly Benchmark, version 0.9. https://github.com/waico/SKAB.
25. SCANIA Component X dataset. Swedish National Data Service, record 2024-34/3, CC BY 4.0.
26. OS-Metrics dataset, version 1.0. Zenodo. doi:10.5281/zenodo.17981352.
27. Bechhoefer, E. (2013). “A Data-Driven Approach to Predictive Maintenance of Industrial Equipment.” MFPT Fault Data Sets. https://www.mfpt.org/fault-data-sets/.
28. Case Western Reserve University Bearing Data Center. Source 97.mat · [Metadata Manifest].
29. Hojjati, H., Roth, C., Woods, R., Sills, K., and Armanfard, N. (2026). “EngineAD: A Real-World Vehicle Engine Anomaly Detection Dataset.” arXiv:2603.25955. Data: version 2.2, doi:10.5683/SP3/TX13P1, CC BY-NC 4.0.

30. Su, Y., Zhao, Y., Niu, C., Liu, R., Sun, W., and Pei, D. (2019). “Robust Anomaly Detection for Multivariate Time Series through Stochastic Recurrent Neural Network.” *Proc. KDD 2019*, 2828–2837. Data (Server Machine Dataset): https://github.com/NetManAIOps/OmniAnomaly.